Really, a small thing:

Big and small; this and that — For every point of light visible in the night sky, there are billions both behind and surrounding it, invisible to the naked eye. The space between the earth and the moon is big enough to fit every other planet in the solar system shoulder to shoulder with almost five thousand miles to spare—yet the sun still dwarfs that distance. The Planck length is the smallest unit of measurement humans have postulated yet it is still almost ten billion times smaller than the diameter of a neutron, one of the smallest particles capable of being observed within the confines of light. The number of Planck lengths that span a human brain cell is approximately the same number of brain cells that span the known universe, give or take a few hundred trillion depending on how you're feeling. The "size" of subatomic particles is only roughly estimated because they cannot be "seen" in the traditional sense; instead, they are measured by how energy is manipulated by their presence—which is like measuring the size of a person by how their voice displaces the air around them in a three-kilometer sphere. To some extent, that's the same way the size of everything is measured. Quarks are capable of being detected in experiments with particle accelerators that cover miles. The world's largest particle accelerator has unveiled the smallest known things. When two quantum particles are next to each other, they become *entangled*: a phenomenon wherein it is impossible to describe them independent of one another—how human! After 2,500 years of "extremely productive work" in the field of linguistics, what we have to show for it is that we still have no cohesive idea for what language is, what a word is, what a dialect is or how babies start talking. The twenty-first and twenty-second centuries are presumed to consist of the greatest number of existential risks humankind will ever face because we will have had (or have already had) the greatest number of technologies capable of destroying ourselves yet we will remain incapable of surmounting threats like climate change, asteroid impact or "human factors." Once humans shed the husk of their own humanness, smooth sailing! *Inflation* refers to the rise in prices that occurs after the value of money decreases—is it so unwise to draw attention to the *cause* of a given phenomenon rather than its *symptoms*? The first word on the United States page in the CIA Global Factbook is "Britain." *Discreet,* meaning *secretive* or *prudent,* and *Discrete,* meaning *distinct,* can be used to illustrate two opposing states of something. Memetically, humans are beyond categorization or earnest comprehension though the human genome is well-known—22,300 protein-coding genes.

One of its findings? the billions of human differences are attributed to 1% of genetic variation—or, rather, 99% genetic singularity.

It's clear out tonight, the air just seems so quiet. Things are starting to feel a little more diminished; shiny things always lose their pallor. Not even the moon is out yet. Just the glow of some stars, the company of Betelgeuse and the Pleiades—mistresses. Fields are never so barren, centers never so fragile, cores never so dim. Poised, motionless in the glow of thinning fog—faltering fields. They grow, or perhaps it's my gaze which shrinks, zooms, focuses. Dev calls to make plans for the weekend. Whatever phenomenon it is, pinprick holes are no longer just points for insipid echoes to briefly pass by, but doves, pigeons, persons—signs of difference, dialectical variation. We grew up together, for twenty years we've known each other, for a twentieth of a greater, grander life we've drifted apart, less than a blink of some cosmic eye; the timbre of our voices has shifted and now, on a pre-winter, almost apocalyptic evening, we make plans to spend a night or two in the vicinity of a home where you can still smell the wildflowers on your porch, beyond the line blurring domesticity and doom. Just a night… or two.

The greatest empire — Dinosaurs ruled the earth as the dominant terrestrial vertebrate species for over 150 million years. They survived through three geological epochs. First appearing, if you're being conservative, 233 million years ago and likely dying 66 million years ago to either the Chicxulub impact event or a large volcanic event leading to a runaway global climate disaster that occurred in India—probably both. They declared their independence from and established their global dominance over the Triassic-Jurassic extinction event 201.3 million years ago. Descendants of theirs exist on almost every continent in the world. To our knowledge, they never irrigated their crops, nor established trade routes, nor expanded their empires, nor did they elect a fascist dictator. There is no evidence of their dominant religions, their art or their culture. They consolidated their power primarily in small, feudal, mostly familial kinships where they would vie for the best local resources and available land as these were scarce and they had no way of stratifying or producing more; they provided care for their young but there is no knowledge on whether they did the same for their elders. They were incredibly diverse with not only thousands of races, but over a thousand distinct species. Interspecies relations are presumed to have been

fierce and competitive, but there is no data on what interracial relations would have looked like. It is also unknown if they were at all cognizant of race, gender or class. They lived in all known climates—with the exception of the arctic—and they thrived in every known terrain. They held every single one the world's natural resources and precious metals for their entire 150-million-year rule. Inflation was at 0% for the entirety of their existence. Their population remained young with few elders; however, the age ratios are unreliable since there is no census data. Their mortality rate is presumed to have been staggering and in the last days of their empire, it reached 100%—that is to say, there is nothing particularly notable about it. There is no known data on their fertility rate. They had an annual GDP of 0 USD. They lived without sanitation. Their literacy rate is unknown. Their unemployment rate was 100%. They suffered greatly, even mortally, from climate disaster. They had no organized militia. There is no known data on their relationship with capitalism, socialism or communism. It is unknown if they would have complied with or assisted in US interests.

Sometimes, we feel hopelessly defiant to holding something, keeping it once, forever. Something to hold gains greasy edges; they slip through our fingers; subjects are lost to the background. When I was ten, I was convinced I would grow up to become an astronomer. I could envision myself peering into the farthest depths of space and uncover whatever lay behind the veil, the obscurity, the darkness. I'm still, for all intents and purposes, petrified of the dark… I'm still. But somewhere between then and now, the moment at which I look back almost 15 years later, I can't help but wonder what changed—or maybe what didn't is the better question. It feels like when change is happening nothing's really happening, but fast forward a few short months and nothing is recognizable… Sometimes, it's immediate, like an apocalyptic storm that sublimates the crags and water in the Gulf, an impact event, a catastrophe. I sometimes think about early cartographers—at which point is their job done? I guess they never really finish their map, they just abandon it—and waves erode the shores. Sometimes. Sometimes.

How "great" we are — Modern humans have only been around for 300,000 years. For every one year that humans have existed, dinosaurs did for almost 430. When we talk about biological and geological epochs, we often measure things logarithmically—the same way the universe seems

to order things of scale. One often has a day where they momentarily become lucid to how much they've changed over the last couple years. Or five. Or ten. And they wonder how different they'll be in another year. Or five. Or ten. They are increasingly unsure, as time goes on, if it is appropriate to consider themselves the same person in passing. And they look down to their arm, trace the follicles that dot their limbs, mapping lines, reaching towards space and they drop back to their place of solace, respite, quiet. They see where they are going, where they're physically going, and they feel like they've just logged onto a stranger's save game. *The world?* They wonder, *what will Eurasia, the world's largest continuous landmass, look like in 150 million years? If each of the 8 billion living human beings gathered there and displaced a single handful of earth by 2 millimeters every year and continued this for 150 million years—what would the planet look like?*

I looked outside my window the other day. This winter's been an unseasonably warm one—like most winters seem to be these days. The snow had just come a few days before. And now, a drift is covering what holes were cast into it. Whispers dispersed into the air like pollen, like sand on dunes, like the drift. Flurries fall and swirl and rise—rotational symmetry, the water cycle, history, memory.

Legacies — it can take microseconds to have a thought and milliseconds to speak. The earliest known written word has seen the rise and fall of almost every organized empire in history yet is less than a fiftieth of the age of those who wrote it. The entropy contained in the displacement of air by One's vocal cords persists for hundreds of billions of years—practically eternity. Eternity encompasses a very small psychological space, it is only a few handfuls of earth, yet "being here, now" is what people repeat to themselves when they're with their best friends and don't want to dread the end of their moment. Nobody ever seems to agree on what the "cause" is, but the "effects" are always clearly evident. Don't even get started on the "implications." An empty galaxy of 100 billion stars is less valuable than a single embryo nestled in the hands of a cold, starless planet. We are taught to take meaning out of art and treasure it, value it—but does it not just add entropy? is it not just the displacement of air by One's vocal cords? a thought? The

cause of art may be muddled, the implications more so, but are the effects, immediate or ultimate, not always beautiful and infinitely expansive? recurring? recursive? even dialectical?

What happened when? — Eta Carinae is a binary-stellar complex and nebula in the milky way galaxy. One of the brightest and most active objects in the night sky, these conjoined stars are an estimated 5 million times more luminous than the sun and 7,500 light years away. They went through what is often referred to as a "great eruption" that began in 1837, the same year that an 18-year-old Queen Victoria acceded the throne and a 25-year-old Charles Dickens published *Oliver Twist*. Over the next 45 years, it fluctuated and oscillated in brightness to a degree barely noticeable to the average observer. This is not the first time this has happened. Later observations have suggested that, because of the surrounding nebulous cloud, the system went through a similar period of change around or during the thirteenth century, the same period that the Kamakura Shogunate established dominance over Japan ending in civil war near the end of the century. Roughly the same time Europe was dying from the plague. Because of the afterbirth of a different star—the last burst of passion of Eta´s parent or parents, the embryo surrounding it, spanning light years, containing buds that will one day bloom into stars and planets after being cast further off into space—the stars are obscured by a pair of glowing orbs, a three-dimensional nephroid. This thing, the Homunculus Nebula, makes it impossible to observe the stars directly; so, they are only observed through the effects they have on the space around them. One or both stars are expected to experience supernovae in the "near astronomical future" (one million years) (a really really long time for any reasonable person). If that happened at this moment, we would be closer in history to the beginning of the series of empires that ruled Egypt than we would be to the point at which we would observe it.

Dev comes to visit. We go out for a night. We eat. We drink. New-adult kid stuff. The next morning, we walk through the city and I tell them all I know about it. I tell them how they'll love it. I tell them how I would *love it. It isn't home; I tell them I miss them. It's new; it's growing; it's becoming who we ought to be. It's forgetting. The fields are fading, the shrubbery withering, wilting. We look around at the bluffs, the valleys, the lithe curves of erosion, of time, of change. Rocks and canopies, sparkles and magic hang overhead, the sky behind starting to get darker, a*

soft trickle of water keeps an eternal whisper, a hum—mistakes and echoes from a sublime god, perhaps, the pulse that steadily buzzing bees keep on the environment—occasional breezes rest their boughs across what used to be summer leaves; our conversations break what stillnesses might perturb the chaos. The universe. The oceanic current. There are rocks here—I can't help but think they're ancient, profoundly ancient, as all are—even the tiniest of sediments displaced by our bare feet as we kick and frolic in the streams and shores back home, maybe hopefully here. Even the water, crystallized shoe-molds in a bed. Our bed. We're new people. We're new. The current sweeps us away. We're new yet again.

What are Eta? — All stars are made up of hydrogen and helium clumped together in such staggering mass that they become harbors for planets; they warp the very fabric of space-time and the two lightest elements in the universe become among the heaviest objects in existence. (Objects? Can they be called objects? They become a form of matter that seems to transcend the very concept of objecthood that such primitive creatures that would go so far as to section off parts of the universe by such a thing as trivial as "objecthood" use to grasp the incomprehensible universe. Is fire an object? Is the flame a different object from the base? The smoke? The heat emanating from it? The translucence that allows one to look through it? The slight stigmatism that forms when the person sitting around the fire stares into the hot coals for too long. No. The star is not separate in its objecthood. Not from the elements ferociously burning in its core, nor the corona surrounding the moon's surface during an eclipse, nor the heat and photons that speed through space to turn to thoughts in One's eyes as they observe from earth. Though a human may be traced by their skin, their voice may still touch everything around them.) The hydrogen and helium in the star undergo changes—they become each other; they drift outward and sink back inward, an elemental stew that brews and circulates in the star's mantle, churning, turning into carbon, nitrogen, oxygen, neon, magnesium and, finally, iron where they smolder like the hot coals at the bottom of the fire pit; gases die in metal, liquid, gaseous plasma-metal; meanwhile, the star is still burning yet beginning to die, diminish. The cross a precipice. Hydrogen and helium are easy to fuse; they bind easily; they require relatively little mass and energy to fuse which, itself, is a process of creating energy which, itself, is a process of creating matter. As time goes on and they run out of H and He to fuse together, they turn to the later, heavier, elements

and continue until they become so massive and the fuel they use to counterbalance the force of their own gravity diminishes. They ultimately collapse in on themselves—some sort of paradox of temporal and volumetric folding and unfolding, unfathomable and nonsensical phenomena, Tuesday for the universe, existence for human beings, fate for all… Every star starts as a blank slate, unknown with regard to which subgroup (gender) (genre) (type) they will inhabit. The conditions handed to them at their conception are crucial in determining which stratification they will remain in for the rest of their lives. A stellar caste casted in a mold of stellar ash. Eta were born with such excesses in early-formation mass that two blue giants were formed, stars that are so bloated in mass and energy that they spin incomprehensibly fast and shine shamelessly. They are so large that the process of stellar collapse is quickened by a factor of over 1,000. They are another statistic in infant mortality rates. Their staggering mass causes the cores to be far hotter than any normal, reasonable star. They are capitalists hell-bent on limitless expansion that they burn all their profits on acquisitions. Instead of living in their main sequence stage for billions of years, they remain there for a mere million or two—still long enough for the evolutionary tribe Hominini to split from the rest of the great apes. While making their lifespan shorter, this also makes them burn far brighter than any other class of star. Anyone who's peered into the Homunculus that imprisons them knows this: they chose the stereotypical life of the deeply troubled yet brilliant artist! Really, they are a small thing.

We walk along and stop at a stream, a river, a shore. Dev and I turn to each other as we begin our utterances:

Aging — It is impossible to know how old or even how massive the stars are. Given the data we have spent the last 150 years observing and gathering, we can estimate that their death will happen in the next million years, a tenth of their lifespan. For comparison, stars like the sun are expected to live for ten billion years. By the time Eta Carinae, one of the brightest objects in the night sky, was born, the dinosaurs had already been dead for 56 million years. When One walks through the woods, their best guestimate for its age is in cracks upon the bark, the height of a given tree, the proximity of human activity. Yet when trees reach their death, almost every event can be mapped. The entropy of wind, rain, snow, the piss of a teenager, the whiplash from the

earth's constant stopping and starting, all of it, can be traced backward in time. When the time comes, when the eternal-recurrence-demon's promise comes true, when the ultimate chaos of all things, our actions' reactions, culminate in the heat-death of the universe, won't our chaotic, entropy-induction engines be discovered then? When Eta grows too big, burns too hot that the core will turn to churning out heavier and heavier elements, won't the nova show the universe what hell brewed within?

Redshift — At this point, we have several telescopes floating around in space—projects that cost about as many dollars as there are brain cells in the average human's dorsolateral and ventromedial prefrontal cortices, which are meant to unveil the deepest secrets of the universe. The newest and most impressive one, the James Webb Space Telescope, is a marvel of modern photographic technology. Its lens is made up of an array of gold-plated hexagonal mirrors adding up to a lens area of over 26 m^2. Its main function is to collect infrared and near-infrared images of the early and distant universe which allows us to look at ancient, deep-space objects with truly magical clarity. Infrared imaging is crucial in this because of the red-shifting effect of distant objects—that is, the relativistic visual phenomenon similar to the Doppler effect wherein objects that are moving away from a given focal point drop in photon energy, and thus electromagnetic frequency, causing them to appear more red. Conversely, approaching objects blueshift and appear slightly blue as they approach. The Doppler effect is this same process but with sound in that the frequency and relative pitch of a sound rises as it approaches and drops as it moves away, commonly observed with an approaching and receding car. The James Webb Space Telescope, with its massive array of gold-plated mirrors, can capture deep space objects at staggering definition and depth. A single grain of sand held at arm's length becomes the entire visual space it plays with, each of those 26 square meters focused on that single grain of sand, sparkling, twinkling in starlight. *I look out to this field—each one a field in itself, its essence, its imperceptibility. Even distant objects, stars whose light takes several millions of years to reach us, are the canvas of Guernica, are entire novels not yet written. Betelgeuse, Eta Carinae, galaxies, nebulae and distant quasars now more acute, distinguished. Sitting in deep space, hardly bound to the clutches of earth's orbital-gravitational field, no longer swinging around its creators in axisymmetric flurries, unbothered by impurities other than the single piece of debris*

which dented one of its reflectors early on in its lifetime, Webb has virtually the whole unburdened universe to play with, to explore, to live in—the Voyager vessels are the only human-made objects with more freedom yet they're still without a massive array of retinae. After 300,000 years of unwritten history and approximately 10,000 years of recorded history, among the most impressive things humans have done is connect single minds to each other and increase our size.

Would we be so lucky as to be out there? Among those insipid figures? When we go to bed at night, we are—floating, flying, even floundering through space; solipsists, certain that existence is an illusion crafted by a nebulous cloud, dust flying through the cosmos—sleep's uncertain and fleetingly tremulous embrace is our awakening. A stage. A small hole in the stage. A falling through the hole and passage into somewhere else. Sediments settling on the shore and tracked before the waves return—or the wind comes, sweeping. Always sweeping. Stars flickering. Inhuman, but no less mystifying.

Entanglement — Pronouns are morphemes that refer to nouns—they are in a state of entanglement. The best analogy I've heard for them is that they're "like pointers in code." When One is writing verbal code (speaking, often thinking), instead of spelling out the entirety of another person's name, for example, they "invoke the pointer" labeled in their lexicon (memory banks, memory) as "he," "she," "they," "we," "I," "you" and "it." They are a tool that makes talking less repetitive, less dry, more *flavorful*. This flavor is often regarded in languages that have a grammatical gender because gender (genre) adds more categorization saying, "this is what kind of thing this is! This is its flavor!" Which is to say they are of a certain morphological shape. Minds like for there to be categories, for there to be a distinct border between things, attitudes, memories, hopes, people. It helps us learn, to make a model of the world that does a better job at being more reflective of reality than before. It helps us remember. Human experience is never disentangled from memory. Even when human beings don't remember, they are forming memories that can either be explicit (remembered) or implicit (habitual, learned). There is *discrete memory* which is contrasted with *discreet memory*. Memory terraforms experience—it gives it flavor, color, even rhythm and idiom. Memories, by virtue of principle,

are never disentangled from time. Time is entangled with experience, memories, hopes, observations, ideas, thoughts, space and stories. When humans use pronouns, they differentiate which pronoun is which by placing them in temporal proximity to that with which they are entangled. Entanglement is a confused mess. Syntax is entangled with time which is entangled with experience which is entangled with memory which is entangled with time which is etc.—how human! It would seem that one of the cornerstones of human intelligence, obvious in language, is the ability to be endlessly recursive—to make the first step for building a stone tool:

Step 1: find a stone tool.

The Present is an illusion because there is nothing One can say *about* the present that is disentangled with observations from past phenomena and filtered through an experience which is entangled with syntax which is entangled with time which is entangled with the rest of language which is etc. Nouns are pronouns for *things* which are pronouns for entangled bits of energy, quarks, elements, compounds, people, nations, societies and the rest.

> *"Do you ever think about the footprints you leave behind?" "What do you mean?" "I mean like when we were on our way out here and when we go back, and forth and again, walking through the dirt and mud and concrete and snow—we left footprints that will stay there for a pretty long time, all things considered, almost like they're telling the story of where we went and who we were, even when it's after we've already ceased." "Yeah, a legacy." "So, you agree?" "Don't you think we're more like footprints in snow? We may not even be around for an afternoon before we're covered up or melt away. Regardless, we all turn to water at the end of the season and become this stream, or the rain, or groundwater that someone will dig up in a few million years should sapient beings be around by then. And I wonder about what happens to the shape of the footprint—does it stay with the water? Does it cease to exist almost as if it never did in the first place? What happens to that specific shape of snow when it melts? Does it just remain as disorder? Chaos?" "I don't know." "Right, neither do I—but isn't it beautiful?*

Why should we want to understand it?"

We decide it would be better for them to stay in town another night rather than try to make it back before dark. We spend some time outside looking up at the starts. I feel drawn to them tonight—I have for quite a while now—so I sit for a couple hours to catalogue what few familiar figures and patterns I see looking back at me. Keys, they're keys into an eternal ether, fresh snow in Valhalla, a dream, an infinite harvest… a joke, maybe. They aren't just shiny—they glow. I think about how all of us do, about how at around 4pm everyday your body is at its peak temperature that it gives off more infrared light than any other time so you are literally glowing brighter. One last thought before I drift off—just whilst my sight is still clear:

Reflection — One steps out of their home near sunset, closer to dusk. They see a vignette of dying sunlight on the horizon, a gradient spanning half the dome under which they will spend their whole life. Above, stars appear as though they're being born from the ash of a dying day. Phoenixes circling an invisible galactic core, tetherballs swinging around a pole with no cord connecting them—untethered balls. They walk to a clearing in the hills behind their country home, wildflowers tickling their nostrils, impassioning memories, bitterness lingering on their tongue. Spring and decay. Soon, a luminous arm spans the ceiling. The walls surrounding the clearing briefly obscure the panorama as they sway in the wind. The speckles and hue above are almost blue. Some oscillate across a spectrum—or seem to. Elsewhere, obscurity. They look into the speckled sky—snowflakes on a silent winter morning, stars pepper the air but with inverted colors. In both cases, silence. Their eyes trace the constellations that confuse the night sky, like blueprints to non-architects, eventually drifting to familiar patterns, eventually to a familiar object, Eta Carinae. Eta was formed by parameters outside of their control and the rest of their existence was determined by that. Like all *things*, at an unknown moment they just *were*. Theoretically, the precise moment at which the first hydrogen atoms fused in what would become the core of Eta Carinae can be pinpointed, but the hydrogen atom that made it a star, then two—the point at which the mass-accumulation provided the necessary conditions for them to start … which grain of sand makes it a mountain? which drop of water makes it an ocean? which genetic base-pair makes it a human? which layer of inter-dimensional disconnection makes it a pronoun? What is our ultimate legacy? What happens when we age? Eta is so young

that they are among the youngest 0.1% of stars and yet they will likely still die before 99% of the remaining 99.9%. Like us, they were created without evident *cause,* without an encyclopedia of the history of the universe loaded into their mind, they knew less about the dinosaurs than we do, and the dinosaurs still lived fifteen years for every year Eta have. Yet like all *things*, they are made of impossibly elusive particles, are, themselves, impossibly elusive particles and add to a legacy of chaos that persists for trillions of years only to, potentially, do it all over again. They are a confused mess. They are two mirrors pointed toward each other, creating a hallway that extends into infinity with a little bit of light dying in each iteration, each point of convergence, each sign of difference in a reflection—a mirrored reflection. *Human.*

Is this what we're bound to? How can I know where only we two are placed? Some clouds move in and obscure my view. Morning comes.